

Outline

- 1) Score function
- 2) Fisher information
- 3) Cramér-Rao Lower Bound
- 4) Examples

Score function

Assume $\mathcal{P}$ has densities p_θ wrt μ , $\Theta \subseteq \mathbb{R}^d$
Common support: $\{x: p_\theta(x) > 0\}$ same $\forall \theta$

Log-likelihood: $l(\theta; x) = \log p_\theta(x)$,

Thought of as random function of θ

$l(\cdot; x) - l(\theta_0; x)$ minimal suff for any fixed $\theta_0 \in \Theta$

Asymptotic picture: What if X highly informative?

Easy to rule out P_θ unless $\theta \approx \theta_0$ (true value)

$\Rightarrow$ "effective model" is $\{P_{\theta_0 + \eta} : \eta \approx 0\}$

$\Rightarrow \underbrace{l(\theta_0 + \eta; x) - l(\theta_0; x)}_{\text{minimal sufficient}} \approx \underbrace{\eta' \nabla_{\theta} l(\theta_0; x)}_{\text{near-sufficient?}}$

Def The score at θ_0 is $S_{\theta_0}(x) = \nabla_{\theta} l(\theta_0; x)$

NB: only a statistic if we specify θ_0 (which nbhd)

Key role in asymptotic statistics

(Rough)

Idea: $p_{\theta_0 + \eta}(x) = e^{l(\theta_0 + \eta; x)} \approx e^{\eta' S_{\theta_0}(x) + \underbrace{O(\|\eta\|^2)}_{\text{important term}}} p_{\theta_0}(x)$

Differential Identities (assuming enough regularity)

$$(*) \quad 1 = \int_{\mathcal{X}} e^{\ell(\theta; x)} d\mu(x)$$

$$\frac{\partial}{\partial \theta_j} : \quad 0 = \int \frac{\partial}{\partial \theta_j} \ell(\theta; x) e^{\ell(\theta; x)} d\mu(x)$$

$$\Rightarrow \mathbb{E}_{\theta} [\nabla \ell(\theta; x)] = 0$$

↑ needs to be same θ

$$\begin{aligned} \frac{\partial}{\partial \theta_k} : \quad 0 &= \int \left(\frac{\partial^2 \ell}{\partial \theta_j \partial \theta_k} + \frac{\partial \ell}{\partial \theta_j} \cdot \frac{\partial \ell}{\partial \theta_k} \right) e^{\ell} d\mu \\ &= \mathbb{E}_{\theta} \left[\frac{\partial^2 \ell}{\partial \theta_j \partial \theta_k} \right] + \mathbb{E}_{\theta} \left[\frac{\partial \ell}{\partial \theta_j} \frac{\partial \ell}{\partial \theta_k} \right] \end{aligned}$$

$$\Rightarrow \text{Var}_{\theta} [\nabla \ell(\theta; x)] = \mathbb{E}_{\theta} [-\nabla^2 \ell(\theta; x)]$$

← same θ →

Def $J(\theta) = \text{Var}_{\theta}(S_{\theta}(x))$ called Fisher information

Central quantity in asymptotic statistics

Score in i.i.d. sample

Assume $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} p_{\theta}^{(1)}(x)$ $\theta \in \Theta \subseteq \mathbb{R}^d$

p_{θ} "regular": common support, $\mathbb{E}_{\theta} \|\nabla \ell(\theta; x)\|^2 < \infty$

$$X \sim p_{\theta}(x) = \prod_i p_{\theta}^{(1)}(x_i)$$

$$\text{Let } \ell_1(\theta; x_i) = \log p_{\theta}^{(1)}(x_i),$$

$$\ell(\theta; x) = \sum_i \ell_1(\theta; x_i)$$

$$S_{\theta}(x) = \nabla \ell(\theta; x)$$

$$= \sum_i \nabla \ell_1(\theta; x_i) = \sum_i \overbrace{S_{\theta}^{(1)}(x_i)}^{\text{univariate score}}$$

$$J(\theta) = \sum_i \text{Var}(S_{\theta}^{(1)}(x_i)) = n \overbrace{J_1(\theta)}^{\text{univariate Fisher info}}$$

Preview of asymptotics

$$\text{Note } \frac{1}{n} \nabla^2 \ell(\theta; x) = \frac{1}{n} \sum_i \nabla^2 \ell(\theta; x_i) \xrightarrow{\text{LLN}} -J_1(\theta)$$

Taylor expansion (d=1)

$$\begin{aligned} \ell(\theta_0 + \eta; x) - \ell(\theta_0; x) &\approx \eta S_{\theta_0}(x) + \frac{\ddot{\ell}(\theta; \eta)}{2} \eta^2 \\ &\approx \eta S_{\theta_0}(x) - \frac{n J_1(\theta_0)}{2} \cdot \eta^2 \end{aligned}$$

Cramér-Rao Lower Bound (aka. Info L.B.)

Suppose $\delta(x)$ unbiased for $g(\theta) \in \mathbb{R}$

$$(*) \quad g(\theta) = \int \delta(x) e^{\ell(\theta; x)} d\mu(x)$$

$$\frac{\partial}{\partial \theta}; \quad \frac{\partial}{\partial \theta} g(\theta) = \int \delta(x) \cdot \frac{\partial}{\partial \theta} \ell(\theta; x) e^{\ell(\theta; x)} d\mu(x)$$

$$\begin{aligned} \nabla g(\theta) &= \mathbb{E}_{\theta} [\delta(x) \cdot S_{\theta}(x)] \\ &= \text{Cov}_{\theta}(\delta(x), S_{\theta}(x)) \end{aligned}$$

Leads to variance lower bound:

$$(d=1): \quad \text{Corr}_{\theta}^2(\delta, S_{\theta}) = \frac{\text{Cov}_{\theta}^2(\delta, S_{\theta})}{\text{Var}_{\theta}(\delta) \text{Var}_{\theta}(S_{\theta})} = \frac{\dot{g}(\theta)^2}{\text{Var}_{\theta}(\delta) J(\theta)}$$

$$\text{Rearrange:} \quad \text{Var}_{\theta}(\delta) = \frac{\dot{g}(\theta)^2}{J(\theta)} \underbrace{\big/ \text{Corr}_{\theta}^2(\delta, S_{\theta})}_{\leq 1}$$

CRLB

$$\geq \dot{g}(\theta)^2 / J(\theta)$$

Interpretation No unbiased estimator has $\text{Var} < \frac{\dot{g}(\theta)^2}{J(\theta)}$

$$\text{If } g(\theta) = \theta, \quad \text{Var}_{\theta}(\delta) \geq \frac{1}{J(\theta)}$$

$$= \frac{1}{n J_1(\theta)} \quad \text{for i.i.d. (sd} \asymp n^{-1/2})$$

Multivariate CRLB

For $\theta \in \mathbb{R}^d$, $g(\theta) \in \mathbb{R}$:

$$\text{Var}_{\theta}(\delta) \geq \nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta)$$

Proof any $a \in \mathbb{R}^d$, $\text{Cov}_{\theta}(\delta, a' S_{\theta}) = a' \nabla g(\theta)$

$$\Rightarrow \text{Var}_{\theta}(\delta) \geq \frac{\text{Cov}_{\theta}(\delta, a' S_{\theta})^2}{\text{Var}_{\theta}(a' S_{\theta})} = \frac{a' \nabla g \nabla g' a}{a' J(\theta) a}$$

Rayleigh quotient, maximized by $a^* = J(\theta)^{-1} \nabla g(\theta)$

$$\Rightarrow \text{Var}_{\theta}(\delta) \geq \nabla g(\theta)' J(\theta)^{-1} \nabla g(\theta) \quad \square$$

Score in Exp. Family

$$\underline{\text{Ex}} \quad p_{\eta}(x) = e^{\eta' T(x) - A(\eta)} h(x)$$

$$\ell(\eta; x) = \eta' T(x) - A(\eta) + \log h(x)$$

Score :

$$\nabla \ell(\eta; x) = T(x) - \nabla A(\eta)$$

$$\Rightarrow S_{\eta}(x) = T(x) - \mathbb{E}_{\eta} T(x)$$

Fisher info (1)

$$\text{Var}_{\eta}(S_{\eta}(x)) = \text{Var}_{\eta}(T(x)) = \nabla^2 A(\eta)$$

Fisher info (2)

$$\nabla^2 \ell(\eta; x) = -\nabla^2 A(\eta)$$

$$\mathbb{E}_{\eta}[-\nabla^2 \ell(\eta; x)] = \nabla^2 A(\eta) \quad \checkmark$$

Curved Exp. Family

$$\underline{\text{Ex}} \quad p_{\theta}(x) = e^{\eta(\theta)'T(x) - A(\eta(\theta))} h(x) \quad \begin{array}{l} \theta \in \mathbb{R} \\ \eta(\theta) \in \mathbb{R}^d \end{array}$$

$$\ell(\theta; x) = \eta(\theta)'T(x) - A(\eta(\theta)) + \log h(x)$$

Score :

$$\dot{\ell}(\theta; x) = \dot{\eta}(\theta)'T(x) - \dot{\eta}(\theta)'\nabla A(\eta(\theta))$$

$$\Rightarrow S_{\theta}(x) = \dot{\eta}(\theta)'(T(x) - \mathbb{E}_{\theta}T(x))$$

Fisher info (1)

$$\text{Var}_{\theta}(S_{\theta}(x)) = \text{Var}_{\theta}(S_{\theta}(x)) = \dot{\eta}(\theta)'\nabla^2 A(\eta(\theta))\dot{\eta}(\theta)$$

Fisher info (2)

$$\ddot{\ell}(\theta; x) = \ddot{\eta}(\theta)'(T(x) - \mathbb{E}_{\theta}T(x)) - \dot{\eta}(\theta)'\nabla^2 A(\eta(\theta))\dot{\eta}(\theta)$$

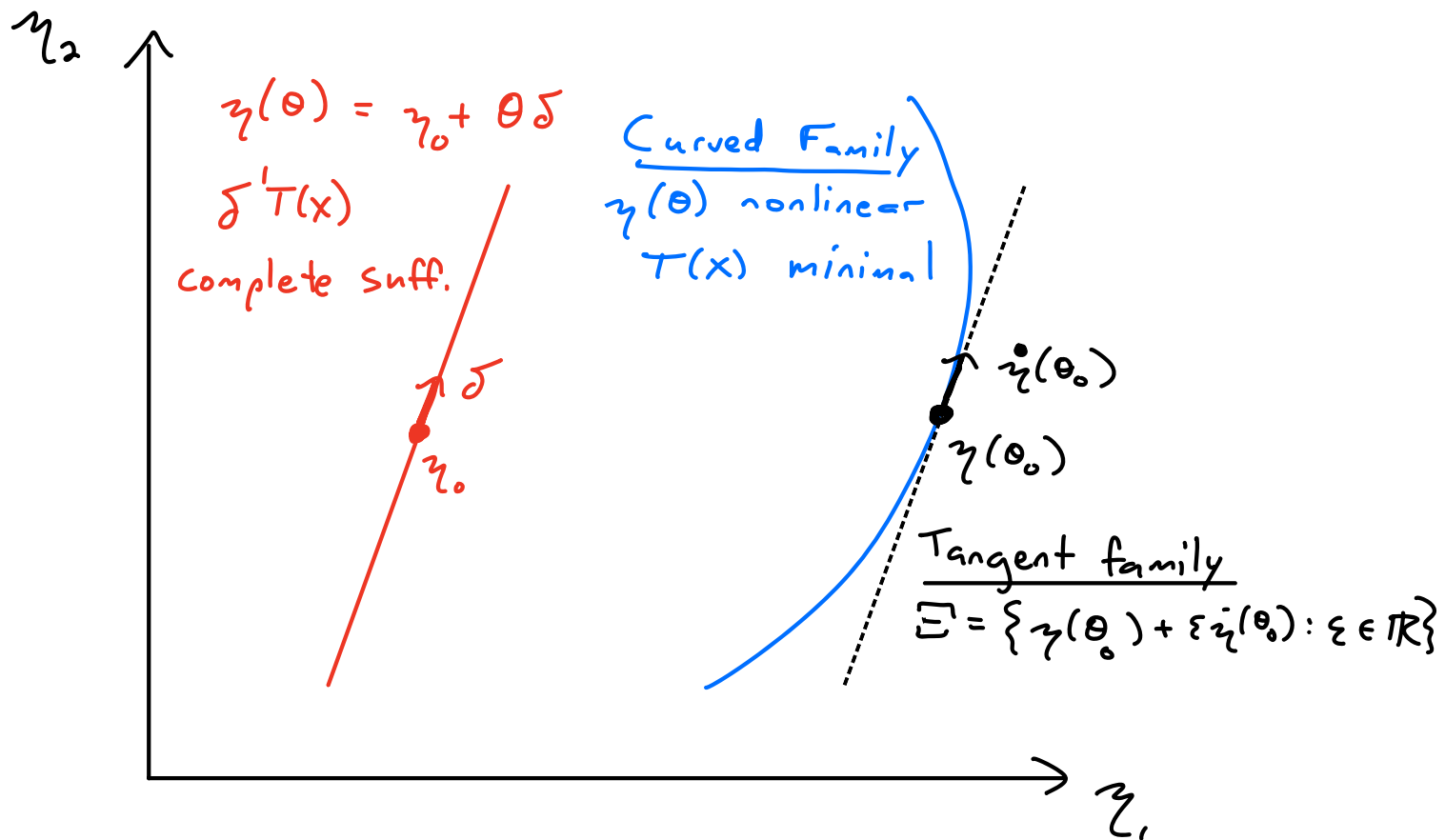
$$\mathbb{E}_{\theta}[-\ddot{\ell}(\theta; x)] = \dot{\eta}(\theta)'\nabla^2 A(\eta(\theta))\dot{\eta}(\theta)$$

Motivation: Tangent family

Concrete if $\mathcal{P}$ is curved exp. family

$$p_{\theta}(x) = e^{\eta(\theta)'T(x) - A(\eta(\theta))} h(x) \quad \eta: \mathbb{R} \rightarrow \mathbb{R}^2$$

$$\Xi = \{\eta(\theta) : \theta \in \mathbb{R}\}$$



$$q_{\varepsilon}(x) = e^{(\eta(\theta_0) + \varepsilon \dot{\eta}(\theta_0))' T(x) - A(\dots)} h(x)$$

$$= e^{\underbrace{\varepsilon \dot{\eta}(\theta_0)' (T(x) - \mathbb{E}_{\theta_0} T)}_{S_{\theta_0}(x)} - B(\varepsilon)} k(x)$$

Complete sufficient for tangent family at θ_0

Ex Curved Gaussian family

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N_d(\mu(\theta), I_d)$$

$$\theta \in \mathbb{R}, \mu(\theta) \in \mathbb{R}^d$$
$$I_d = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \in \mathbb{R}^{d \times d}$$

Sufficiency reduction:

$$\bar{X} \sim N_d(\mu(\theta), \frac{1}{n} I_d)$$

$$\rho_{\theta}^{(1)}(x) = (2\pi)^{-d/2} \cdot e^{-\|x - \mu(\theta)\|^2/2}$$

$$\rho_{\theta}(\bar{x}) = \left(\frac{2\pi}{n}\right)^{-d/2} e^{-n \|\bar{x} - \mu(\theta)\|^2/2}$$

$$= \exp \left\{ n \mu(\theta)' \bar{x} - n \frac{\|\mu(\theta)\|^2}{2} \right\} \cdot h(x)$$

$$S_{\theta}(\bar{x}) = \nabla \log \rho_{\theta}(\bar{x}) = n \dot{\mu}(\theta)' (\bar{x} - \mu(\theta))$$

$$J(\theta) = n \underbrace{\|\dot{\mu}(\theta)\|^2}_{\substack{\text{squared rate of} \\ \text{change wrt } \theta}}$$

sample size

Ex: $d=2, \mu(\theta) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \quad \theta \in [0, \pi/2]$

